Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.–3)

PHARMACEUTICAL ENGINEERING

Subject Code : BP-304T

M.Code : 75108

Date of Examination:09-12-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION - A

1. **Write short notes on :**

   a. Write any two objectives of filtration.

   b. What are the applications of centrifugation?

   c. Define size reduction.

   d. Name any two laws governing size reduction.

   e. What is the function of paddles in mixing?

   f. Name two types of heat exchangers.

   g. Mention two types of distillation.

   h. What is meant by molecular distillation?

   i. Give two examples of non-metallic materials used in plant construction.

   j. Write two factors influencing evaporation.

Chandigarh College of Pharmacy LIBRARY Landran (Mohali)

NOV 20

SECTION-B

2. Explain the working principle, construction, uses, merits and demerits of tray dryer and spray dryer.

3. Describe Bernoulli's theorem and its pharmaceutical applications. How are orifice meters and venturimeters used for flow measurement?

4. Explain the various mechanisms of heat transfer. Discuss Fourier's law and its importance in pharmaceutical engineering.

SECTION - C

5. Describe any two methods of size separation.

6. What are the factors affecting mixing of solids?

7. Write a note on the working principle of Sigma blade mixer.

8. Differentiate between flash distillation and steam distillation.

9. Explain the principle and working of centrifugal filters.

10. Describe the applications of fluidized bed dryer.

11. Discuss the types of corrosion in pharmaceutical plants.

12. What is the significance of manometers in flow measurement?

13. Describe the construction and working of filter leaf.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

025